

BUIDL CTC 2026 FALL · DEFI

Kitty

Savings circles where every payment is proven, not promised.

Ethereum stablecoins · settled and credit-scored on Creditcoin · via the Attestcoin Protocol

PROBLEM

Hundreds of millions save in circles nobody can see.

- Chit funds, susu, tandas, chamas, stokvels: ten friends, one pot a month.
- They fail two ways: the treasurer runs, or a member stops paying and nobody outside the group ever knows.
- Ten years of perfect payments builds zero formal credit history.

TRUST TODAY

Who you have to trust

STEP	INFORMAL CIRCLE	“ROSCA ON-CHAIN” APPS	KITTY
Holds the money	Treasurer	One contract on one chain	Escrow vault on Ethereum
Knows who paid	Treasurer’s notebook	Same contract	Attestcoin-proven txs on Creditcoin
Decides the deadline	Treasurer	block.timestamp / admin	Attested source-chain block height
Picks the recipient	Treasurer / lottery	VRF oracle	Fixed order or by proven score
Credit history	None	None	Kitty Score, readable by any lender

SOLUTION

Money on Ethereum. Rules on Creditcoin. Proofs in between.

- **Not an import of DeFi history someone already has.** Kitty creates a primary credit record from savings circles that run today with no bank: every installment, deadline, payout and miss of the circle itself is proven or attested.
- Members pay into a minimal escrow vault on Sepolia.
- The attestor network attests the block; one batch proof covers the whole round.
- KittyLedger verifies through precompile 0x0FD2, decodes and binds every payment, enforces deadlines from 0x0FD3.
- Payouts are proven back. “Paid” is never an operator’s word.
- Any member can prove a round from the browser: the Proof Builder serves CORS, the ledger doesn’t care who submits.

LIVE LOOP · PAY → ATTEST → PROVE → VERIFY → CLOSE → PAY OUT → PROVE BACK

The first circle, live: Delhi Chit Circle



CIRCLES

7

live · Creditcoin CC3 Testnet

PAYMENTS PROVEN

41

95% on time

BATCH PROOFS

13

43 tx through 0x0FD2

TUSD SETTLED

3,350

payouts proven back

ATTESTED SEPOLIA
BLOCK

11697450

latest · 0x0FD3

KITTYLEDGER

0xC2A1...F276

Creditcoin CC3 Testnet ·
Blockscout

● proven on time ● proven late ● paid · proof pending ● missed

— receives this round ✓ already received

3 members · 100 tUSD a round · rotation by Kitty Score · round 3 of 3

What the ledger actually checks

- Batch `verifyAndEmit`: up to 10 txs, one continuity proof, one call for payments pooled across every open circle, preflighted free with the view verify, 24% cheaper than singles on the live 0x0FD2.
- EvmV1Decoder: receipt status 1, exactly one Contributed log, emitter = registered vault, tx.to = vault, tx.from = member, exact amount, current round.
- Query id = keccak(chainKey || height || txIndex), same as ASCBase; per-circle chain key validated with `get_chain_by_key`; vault allowlist keyed by chain.
- Deadlines from attested height: `is_height_attested(chainKey, deadline + 64)`; a miss records the attestation that proved it.
- The dashboard asks 0x0FD3 which attestation covers a payment: `find_lowest_attested_after, get_attestation_bounds`.
- Proven against the live precompile before deploying: a real Sepolia proof → `0x0FD2.verify = true`; tampered bytes and a wrong chain key `revert(pnpm verify:live)`.

src/asc/KittyLedger.sol · recordContributions

one precompile call

```
function recordContributions(
    uint64 chainKey,
    uint64[] calldata heights,
    bytes[] calldata encodedTxs,
    INativeQueryVerifier.MerkleProof[] calldata merkleProofs,
    INativeQueryVerifier.ContinuityProof calldata continuity
) external {
    bytes32[] memory queryIds = _prepareBatch(chainKey, heights, encodedTxs, merkleProofs);
    uint256 n = heights.length;

    bool ok = VERIFIER.verifyAndEmit(chainKey, heights, encodedTxs, merkleProofs, continuity);
    if (!ok) revert ProofRejected();

    uint64 lo = type(uint64).max;
    uint64 hi;
    for (uint256 i; i < n; ++i) {
        processedQueries[queryIds[i]] = true;
        _recordContribution(chainKey, queryIds[i], heights[i], encodedTxs[i]);
        if (heights[i] < lo) lo = heights[i];
        if (heights[i] > hi) hi = heights[i];
    }
    emit BatchVerified(chainKey, lo, hi, n);
}
```

Eight ways to cheat: six decoded rejections, two accepted by design

- Replay → QueryAlreadyProcessed
- Fake vault with a perfect event → WrongEmitter
- Same proof, chain key 3 → WrongChain
- Included but reverted tx → SourceTxFailed
- Paid after the deadline → accepted, flagged late, -20
- Steal the steward's key → NotOperator / OwnableUnauthorizedAccount / RoundStillOpenOnSource / VaultNotTrusted
- Fire the agent → accepted: the ledger checks the proof, not the caller
- Poison the reasoning → three invented figures stripped before display

Reviewed adversarially before submission: only owner-trusted vaults can feed the ledger, nobody can be penalised for a circle they never consented to, a 64-block grace window protects on-time payers from a hostile early close, and pots only go to members who paid this round.

An agent whose only power is proof

LAYER	HOLDS	WHAT IT DOES
1 · The ledger	final say	Verified proof, receipt status 1, trusted emitter, right sender and target, exact amount, current round, unseen query id. Any failure: nothing happens.
2 · Deterministic decisions	timing only	Prove now or wait for a fuller batch? Missing the grace window costs 120 points: urgency beats thrift. Decisions logged.
3 · Cited reasoning	none	Claude explains the log in plain language. Any sentence with an uncited or unverifiable figure is stripped before display. No API key: the deterministic sentence, identical behaviour.

payout 13 Sept, 07:06 UTC

paid circle 5 round 1: 300 tUSD to
0x128aC52048072BeEb5b930Cc5D9F8c91EDE5bEB
A

circleId=5

round=1

recipient=0x128aC520...bEBA

amount_tUSD=300

sepolia 0x48f314d0e4...

confirm 13 Sept, 07:16 UTC

payout for circle 5 round 1 proven back to Creditcoin

circleId=5

round=1

sourceHeight=11694351

creditcoin 0xb6e6091e31... sepolia 0x48f314d0e4...

close 13 Sept, 08:45 UTC

closed circle 5 round 2 — close height 11694784
attested

circleId=5

round=2

closeHeight=11694784

proven=0

members=3

everyoneProven=false

creditcoin 0x68cc0eeacd...

KITTY SCORE → CREDIT

The score is used, not just displayed

- 500 base · +15 on time · −20 late · −120 missed · 300–850.
- KittyCreditLine on Creditcoin underwrites from the score alone: tier A borrows 100% of proven volume, B 50%, C 20%, D nothing.
- A soulbound Kitty Score badge renders live from the ledger, so a lender can read it on any explorer.
- Every input is a proven transaction or an attested deadline. This is the data Creditcoin was built to carry.

PRODUCT

Everything a circle needs, from a browser or a phone

- **Create** a circle in one Creditcoin transaction from `/create`: listed members or an open circle with signed invite links; the chain picker reads `get_supported_chains` from `OxOFD3` and the start height is placed beyond the attested frontier.
- **Join** by invite link (`/join`) or accept membership on the circle page; only consented members can be marked missed.
- **Pay** on Sepolia and **prove** the whole round yourself from the same wallet: attestation bounds, Proof Builder, free preflight, one `verifyAndEmit` call.
- **Telegram**: [@KittyCirclesBot](#) answers `/circle`, `/score` and `/steward` from chain reads, pushes every proof and payout, reminds before deadlines, and opens the dashboard as a Mini App.
- Score dial, rotation wheel, steward decision log and the recorded attack lab, all reading the live ledger.

The first circle has already settled

- KittyLedger, KittyViewer, KittyUSD, KittyCreditLine and KittyBadge deployed to Creditcoin CC3 Testnet on 12 September 2026 (KittyLedger 0xC2A1...F276); its paired Sepolia vault 0xa27e...DA84 is trusted on the ledger; the first circle and all eight attack scenarios ran on the first ledger and vault.
- Delhi Chit Circle, three members, 100 tUSD a round, rotation by score: three Sepolia payments, one batch proof of three verified by the live 0x0FD2 in a single call, an early close, a 300 tUSD payout on Sepolia and its proof back to Creditcoin. Then two more circles: eight payments settled in one cross-circle call, a member proving from the browser, and a real miss closed on the attested deadline.
- Attack scenarios rerun against the real precompile: the forged chain key is rejected by 0x0FD2 itself. Every transaction is in docs/TESTNET_LOG.md; the hosted dashboard, steward log and attack lab read the same chain.

ACTION	CHAIN	TRANSACTION	GAS
deploy KittyLedger · the final ledger, 0xC2A1...F276	Creditcoin	0x21994599...7ac7f5	4,901,416
recordContributions round 0 · the whole round, batch of 3, one 0x0FD2 call	Creditcoin	0xac2a637f...652fe9	888,573
closeRound 0 · everyone paid, early close, rotation by score	Creditcoin	0x53cbb51d...277bbb	354,326
payout round 0 · 300 tUSD to member 0	Sepolia	0xfee30618...7263c3	64,821
confirmPayout round 0 · proven back through 0x0FD2	Creditcoin	0x92344d88...74ba88	386,582
recordContributions · 8 payments from circles 2 and 3 in ONE precompile call	Creditcoin	0xa7310f00...c1f7fa	1,936,724
recordContributions circle 3 round 1 · proven from the browser by the member, no operator	Creditcoin	0x7b8fdaab...8861b5	401,366
closeRound circle 1 round 1 · attested deadline + 64, a real miss with its attestation	Creditcoin	0xef146316...9f01ca	362,992

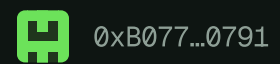
Who arrives, where this goes, how it gets built

Users	Savers who already run chit funds, susu, tandas and chamas. They arrive through the Telegram bot and Mini App, invite links and ordinary stablecoin payments; their history is readable by any Creditcoin lender.
Roadmap	Mainnet circles under chain key 3 with a USDC vault; score-gated circle sizes and seat bidding; pot cover from proven events; defaults feeding the score; lender integrations; payouts through Attestcoin writability.
Execution	100 commits in six days by one builder: contracts, steward, dashboard, bot, documentation, four testnet deployments, 127 linked transactions, a recorded demo. Every problem found live was fixed and logged the same day.
Proven models	A centuries-old savings mechanism, regulated as chit funds in India; the credit-bureau score lenders already price against; a per-round fee priced against the foreman's commission, made cheap by batch proofs.

TEAM

Prashant · solo builder

Foundry · TypeScript · React. Repo: <https://github.com/Prashant-thakur77/Kitty>. Contracts on Sepolia and Creditcoin CC3 Testnet; 162 Foundry tests · 29 agent tests · 8 attack scenarios end to end in CI; verified against the live Ox0FD2.



member 0 · first pot received

KittyLedger 0xC2A1...F276